Reg. No. | | | | | | | | | | |

[14EE63]

SIXTH SEMESTER B. TECH EXAMINATION, JUNE - 2019

POWER SYSTEM PROTECTION

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions, by selecting any two questions from each part.
2. Missing data may be suitably assumed.

PART-A

1. a. Explain difference between isolating switch and load breaking switch. **04**

   b. Explain the following terms related to fuse, **10**

      i. Minimum fusing current

      ii. Rated current

      iii. Prospective current

      iv. Cut-off current

      v. Fusing factor

2. a. Derive an expression for Restriking voltage and Rate of Rise of Restriking Voltage (RRRV). **06**

   b. In a 132 kV system, the inductive reactance and capacitance up to the location of circuit breaker is 3 Ω and 0.015μF respectively. Calculate the following, (Assume frequency 50Hz). **04**

      i. The frequency of transient oscillation

      ii. The maximum value of restriking voltage across the constants of the circuit breaker.

      iii. The maximum value of RRRV.

   c. Discuss briefly problems encountered in DC-circuit breaking. **04**

3. a. Explain in brief puffer type SF6 circuit breaker. **10**

   b. Discuss advantages of air blast circuit breaker over oil circuit breaker. **04**

4. a. With a neat sketch explain vacuum circuit breaker (VCB). **08**

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b. Explain Synthetic testing of a circuit breaker with neat diagram. **06**

PART-B

5. a. State essential qualities of protective relay. **04**

b. Define the following terms. **04**

i. Pick-up Value

ii. Reset Value

c. What is the role of back-up protection? And what are various methods of back-up protection. **06**

6. a. Discuss protection scheme for parallel feeder and protection of ring mains. **08**

b. How directional earth fault relay is energized.? **03**

c. The current rating of a relay is 5A. PSM = 1.5, TMS = 0.4. CT ratio is 400/5, fault current is 6000 A. Determine the operating time of the relay at TMS =1.Operating time at various PSM are, **03**

PSM	2	4	5	8	10	20
Operating time in Seconds	10	5	4	3	2.2	2.4

7. a. Explain operating principle of impedance relay and its operating characteristics. **06**

b. Discuss characteristics of three zone impedance relays with directional unit. **04**

c. Draw the Schematic diagram of microprocessor based over current relay. **04**

8. a. Explain protection against stator inter turn faults of generator. **06**

b. Discuss various faults encountered in transformer and explain percentage differential protection for star – delta connected transformer. **08**